

QINTERN 2026 · FINAL PRESENTATION

Quantum Sentinel

A Rigorous Benchmark of Quantum Neural Networks for Zero-Day
and Adversarial Intrusion Detection in IoT Networks

 Team A

Data Curation & Conformal Guarantee

Amon Koike, Mohammed Owais, Iwo Wojtakajtis

 Team B

Intelligent Feature Selection

Kevin Wong, Maral Mahmoudi Kamelabad

 Team C

Quantum Training & Robustness

Arjun E Naik, La Wun Nannda

Mentors: *Dr. Mohit Sajwan · Dr. Simranjit Singh*

What is Quantum Sentinel?

A reproducible benchmark that asks a single, honest question: **can quantum neural networks detect the network attacks they were never trained on?**

We combine three ingredients into one pipeline:

- 1 Quantum encoding** — network flows become quantum states through a variational circuit
- 2 Prototype fidelity** — each flow is scored by how unlike every known attack it looks
- 3 Conformal calibration** — a distribution-free guarantee on the false-alarm rate

AT A GLANCE

3

IoT intrusion datasets (CIC-IoT2023, BoT-IoT, UNSW-NB15)

3

quantum architectures + parameter-matched baselines

4

evaluation axes: in-distribution, zero-day, noise, adversarial

17

unified features, one shared schema — qubits, seed 42, $\alpha = 0.05$ (the target false-alarm rate)

Five questions we set out to answer

RQ1

Detection

Does the quantum model match strong classical baselines on known attacks?

RQ2

Zero-day

Does conformal calibration hit its target false-alarm rate while keeping recall?

RQ3

Disentanglement

Can it tell a disguised known attack from a genuinely novel one?

RQ4

Robustness

How does it hold up under NISQ noise and FGSM / PGD adversarial attacks?

RQ5

Honesty

Where does quantum NOT help — and what does it cost?

Prior quantum intrusion detection literature suffers from three major flaws:

1

Single-dataset claims

Single-dataset claims on noiseless simulators

→ **QS-Net: 3 datasets, one schema.**

2

No false-alarm guarantee

Hand-tuned thresholds with zero statistical bounds on false alarms

→ **QS-Net: exact finite-sample band.**

3

No adversarial account

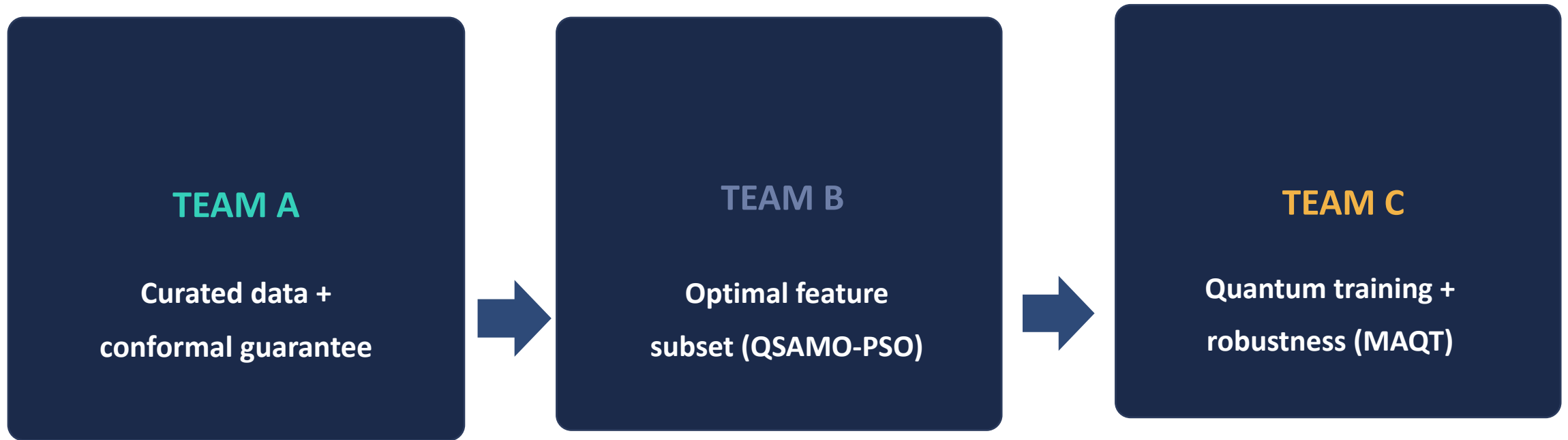
What happens when an attacker disguises a known attack as novel? Untested.

→ **QS-Net: a 4th evaluation axis.**

OUR PROJECT

Our goal wasn't to artificially prove quantum superiority, but to rigorously map where quantum models succeed and where they fail.

Three teams, one pipeline



- Team A:** First to use quantum state fidelity in Conformal Prediction to provide a mathematically guaranteed false-alarm bound ($P(\text{error}) \leq \alpha$) on leak-free zero-day splits.
- Team B:** Developed QSAMO-PSO, a 5-objective optimization algorithm tailored specifically to minimize NISQ quantum circuit depth while preserving zero-day detection.
- Team C:** Introduced Margin-Aware Quantum Training (MAQT) and derived theoretical certified adversarial robustness bounds under depolarizing noise.



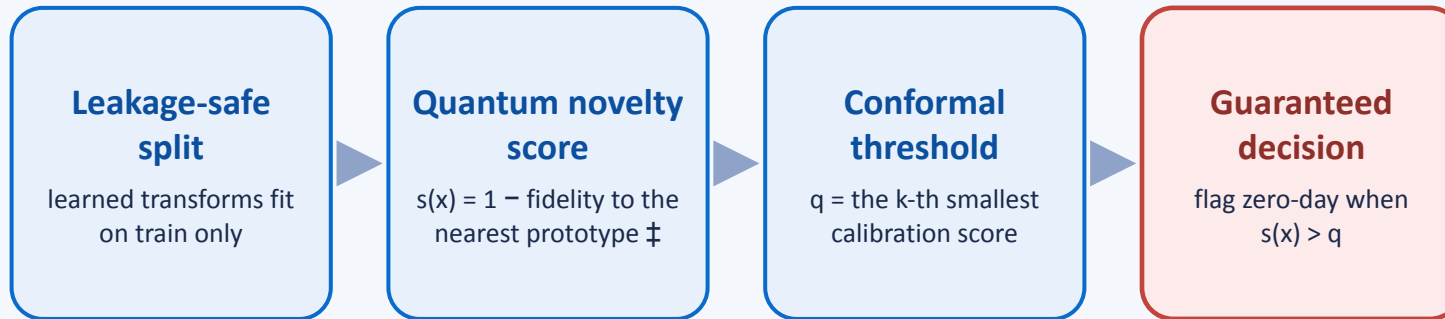
TEAM A

Data Curation & the Conformal Guarantee

Amon Koike · Mohammed Owais · Iwo Wojtakajtis

One guarantee, bolted to a quantum score

FROM A RAW FLOW TO A PROMISED ERROR RATE



$\mathbf{x}$ a network flow · $s(\mathbf{x})$ its novelty score · q the threshold · k fixed by α , never tuned · α the false-alarm rate we ask for

THE GUARANTEE

$P(\text{false alarm}) \leq \alpha$ — at any sample size, for any score function, even a bad one

THE ASSUMPTION, TESTED

Exchangeability can only be falsified: 18 tests clean, then 9 injected breakages — all 9 caught

‡ protocol real, fidelities not yet — the quantum arm is still Team B's placeholder

WHAT IS NEW

- Fidelity-to-prototype as a conformal nonconformity score — novel as far as we can find
- The bound is distribution-free — it holds even when the score is bad
- The gate is the exact Beta-Binomial band, not a rate-under- α check
- The assumption is falsification-tested, not asserted

The data foundation & zero-day partitions

SPLIT FIRST, THEN FIT ON TRAIN ONLY

31 / 4 / 8 known classes



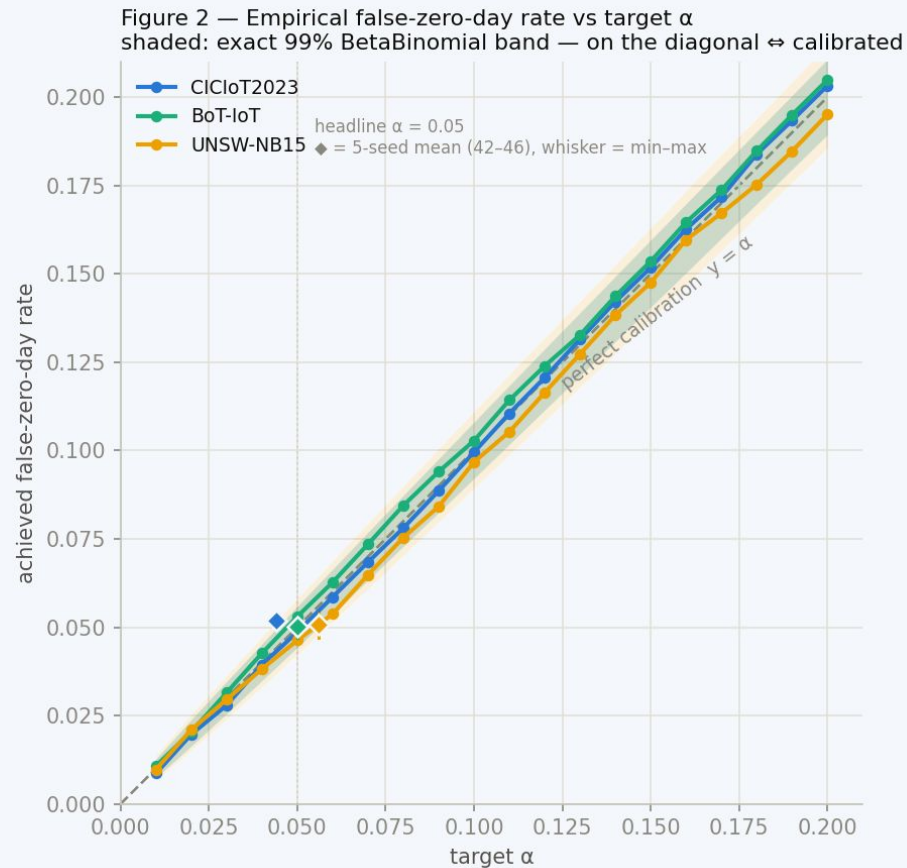
Dataset	Calibration	Test	Zero-day	Held-out family
CIC-IoT2023	18,883	18,883	10,984	Mirai (3 variants)
BoT-IoT	18,591	18,591	683	Theft
UNSW-NB15	10,112	10,086	1,216	Shellcode + Worms

Leakage designed out by ordering, then verified: two-sample AUROC 0.4999 / 0.5038 / 0.5035

WHAT WE BUILT

- Three corpora on one 17-feature schema; two were set aside (TON_IoT and Edge-IIoTset), not solved
- Calibration is known-classes only — zero-day rows there inflate q and void it
- Zero-day is a whole held-out family, never a random subsample
- Leakage designed out by ordering, then verified on the shipped splits

The guarantee holds — and we test it correctly



WHAT IT SHOWS

- Here we show the empirical calibration curve. The dotted line is our target false alarm rate $\alpha = 0.05$, and the shaded region is the exact 99% Beta-Binomial confidence band.
- All three datasets across all 5 random seeds stay strictly inside the band. This proves that our conformal calibration delivers the promised false-alarm bound in practice.

In the figure, “false-zero-day rate” is that same rate — known-class rows wrongly flagged as novel.

How the threshold is set

1

Score the calibration rows

Known classes only. Each row gets one novelty score — 1 minus its fidelity to the nearest known-attack prototype.

2

Sort, then take the k-th

$\hat{q}$ is the k-th smallest of those scores, with $k = \lceil (n+1)(1-\alpha) \rceil$. Nothing is fitted here, and nothing is tuned.

3

Flag anything above $\hat{q}$

Applied unchanged to Test and to Zero-day. A fresh known-class row exceeds $\hat{q}$ with probability at most α .

n calibration scores, sorted low $\rightarrow$ high novelty

$\hat{q}$ sits at the k-th score $\cdot k = \lceil (n+1)(1-\alpha) \rceil$

$1 - \alpha$ of known-class rows stay silent

$\alpha = 5\%$ flagged, by construction

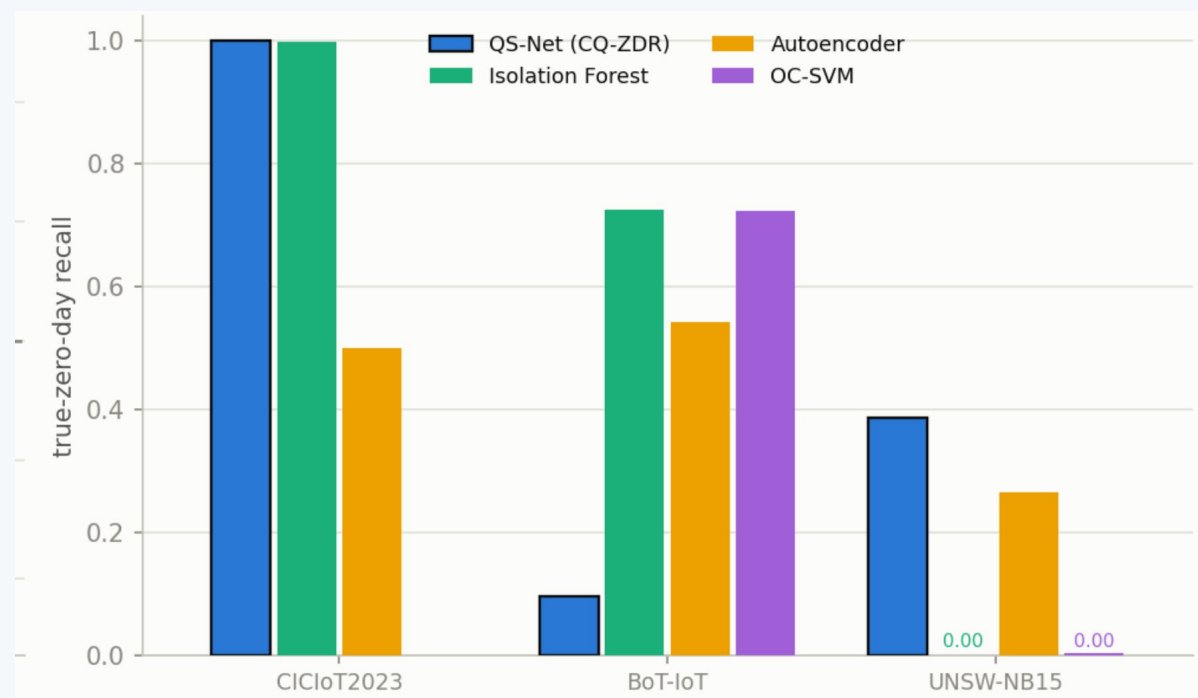
WHY IT HOLDS

Distribution-free and finite-sample: the bound assumes nothing about the score being good — only that calibration and test rows are exchangeable, which we test rather than assert.

Coverage vs recall — the honest comparison

TRUE-ZERO-DAY RECALL AT THE SAME α

Same guarantee for every head — this is where they actually differ



QS-Net is the outlined bar. A numeric label marks a head scoring ≈ 0 ; an empty slot is a head that dataset has no model for.

WHAT SEPARATES THEM

- Thanks to conformal calibration, every model is guaranteed to stay below 5% false alarms—so the only real test is Zero-Day Recall.
- The results are honest:*
- Quantum is not an automatic silver bullet; its advantage depends entirely on the attack geometry. Over to Team B.

Δ RECALL vs BEST CLASSICAL

CIC-IoT2023	+0.002
BoT-IoT	-0.628
UNSW-NB15	+0.120

true-zero-day recall at $\alpha = 0.05$



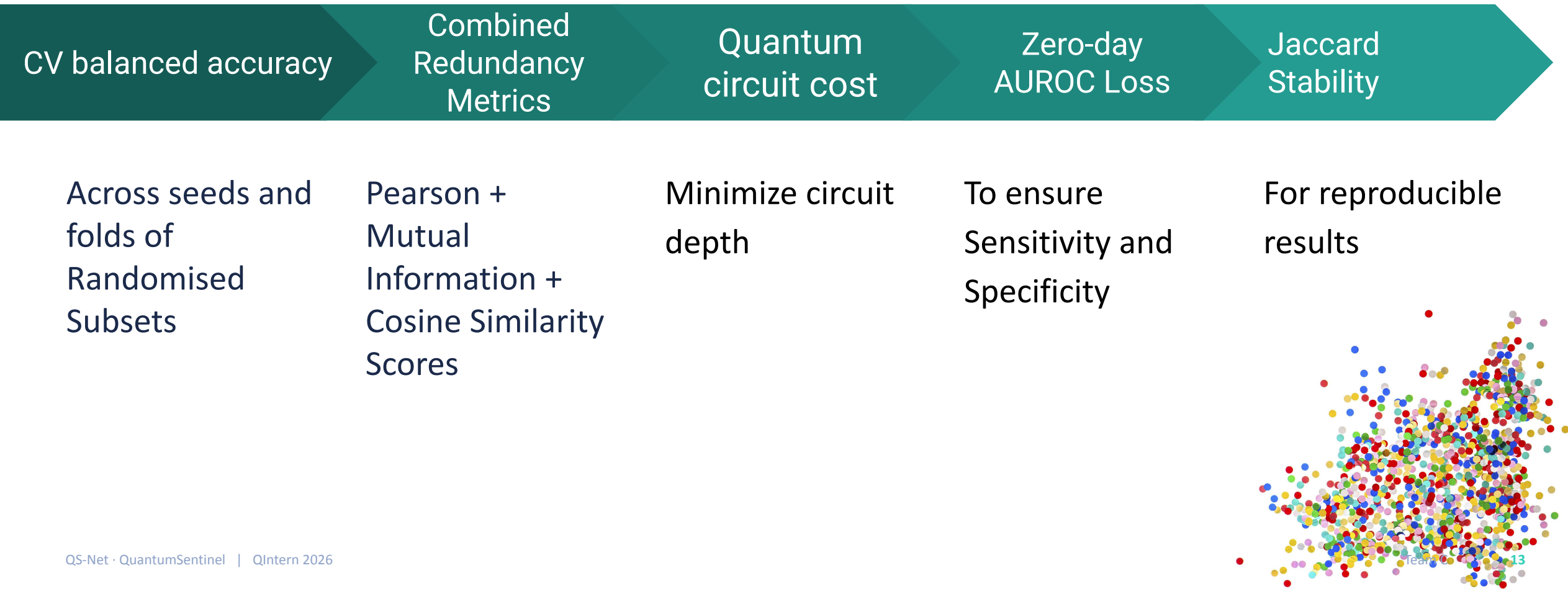
TEAM B

Intelligent Feature Selection

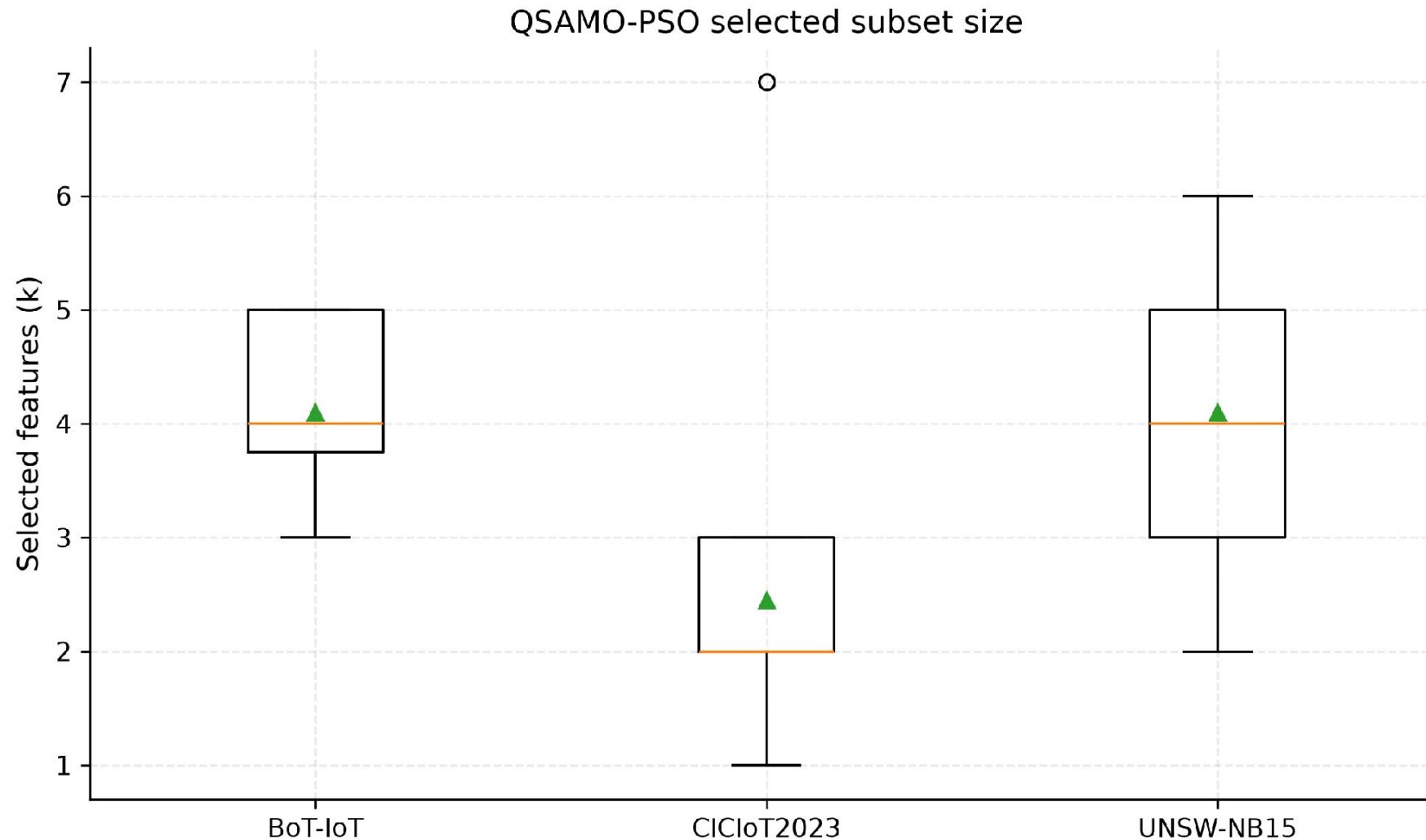
Kevin Wong · Maral Mahmoudi Kamelabad

QSAMO-PSO: multi-objective feature selection

Quantum Self-Adaptive Multi-Objective Particle Swarm Optimization for Feature Selection

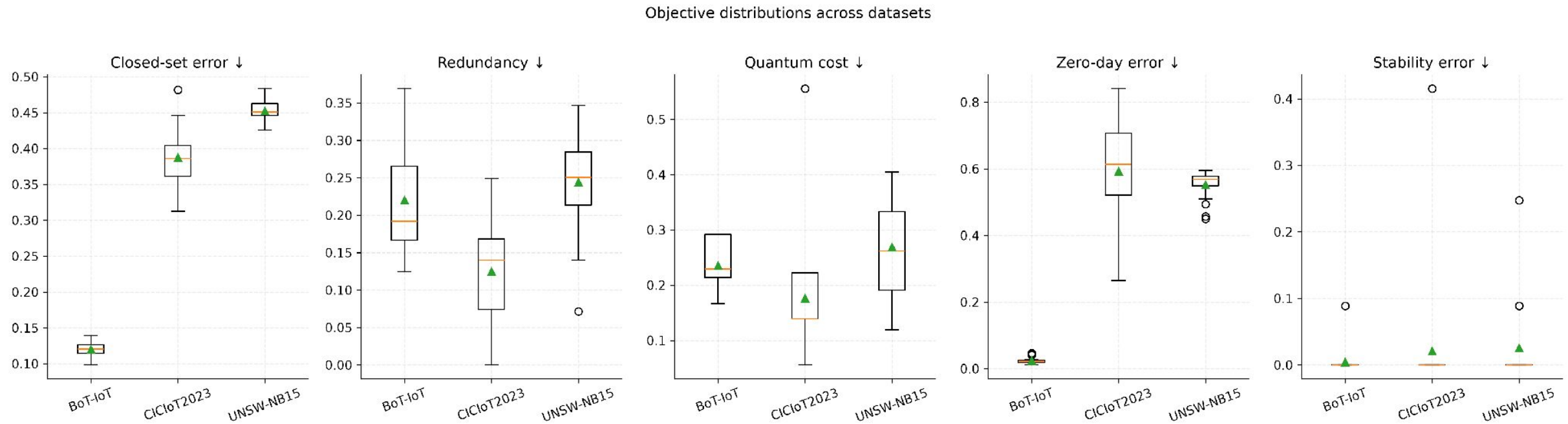


Number of Features Selected Across Datasets



- Starting from 12-17 features
- Across 5 seeds
- 7-k fold CV

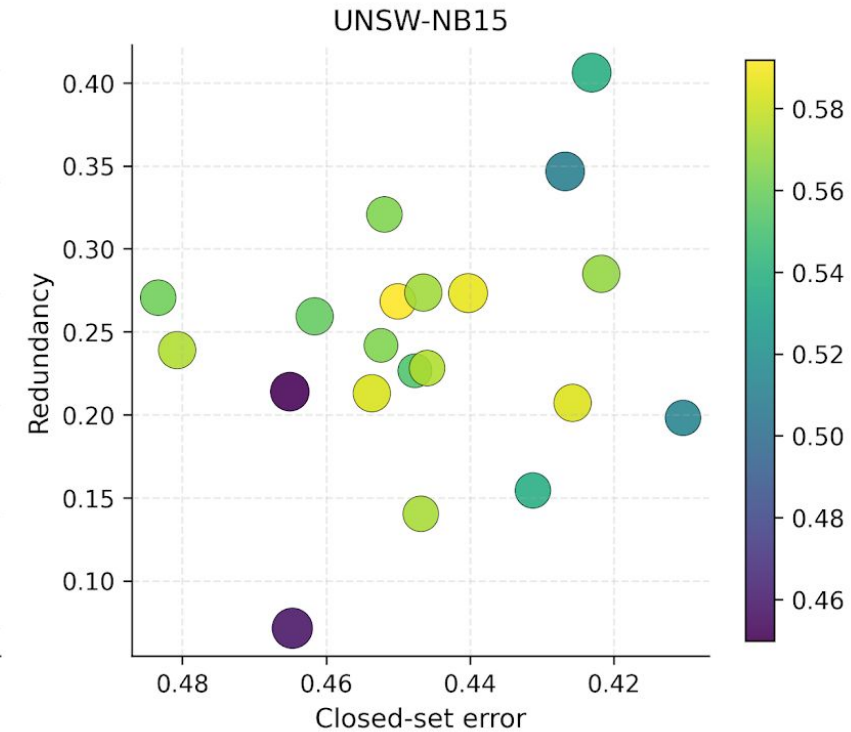
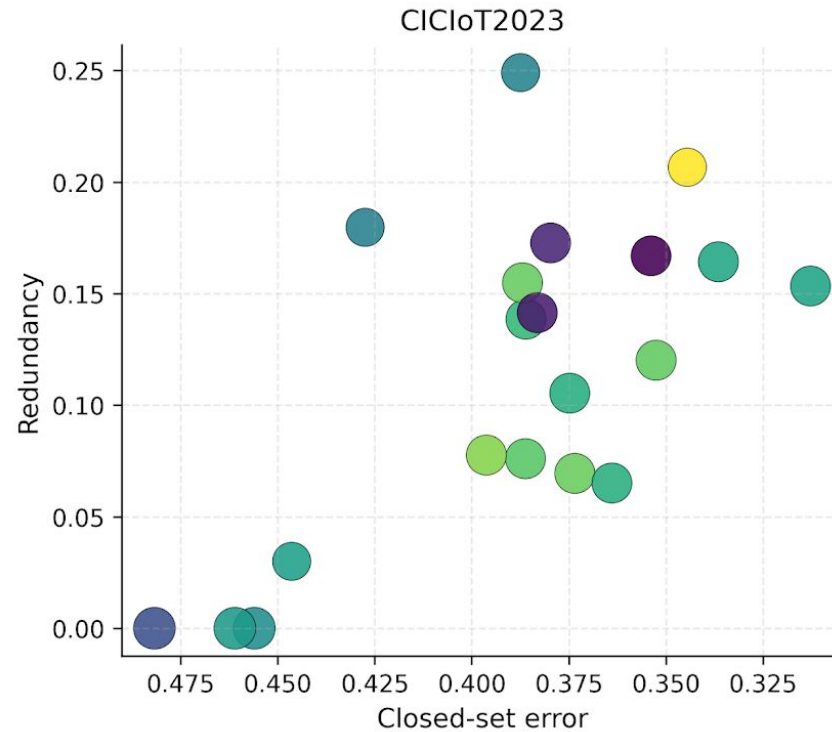
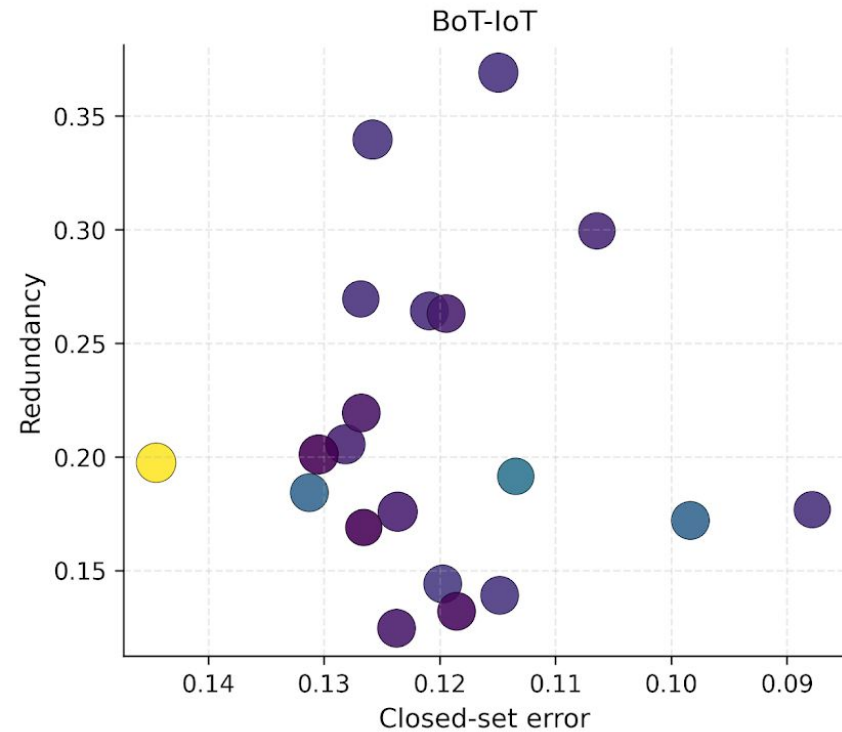
Objective distributions across datasets



Captures key variance in each dataset across every single objective

Pareto front projection

Pareto projection ($x=f_1$, $y=f_2$, $\text{color}=f_4$, $\text{size}\approx 1-f_3$)

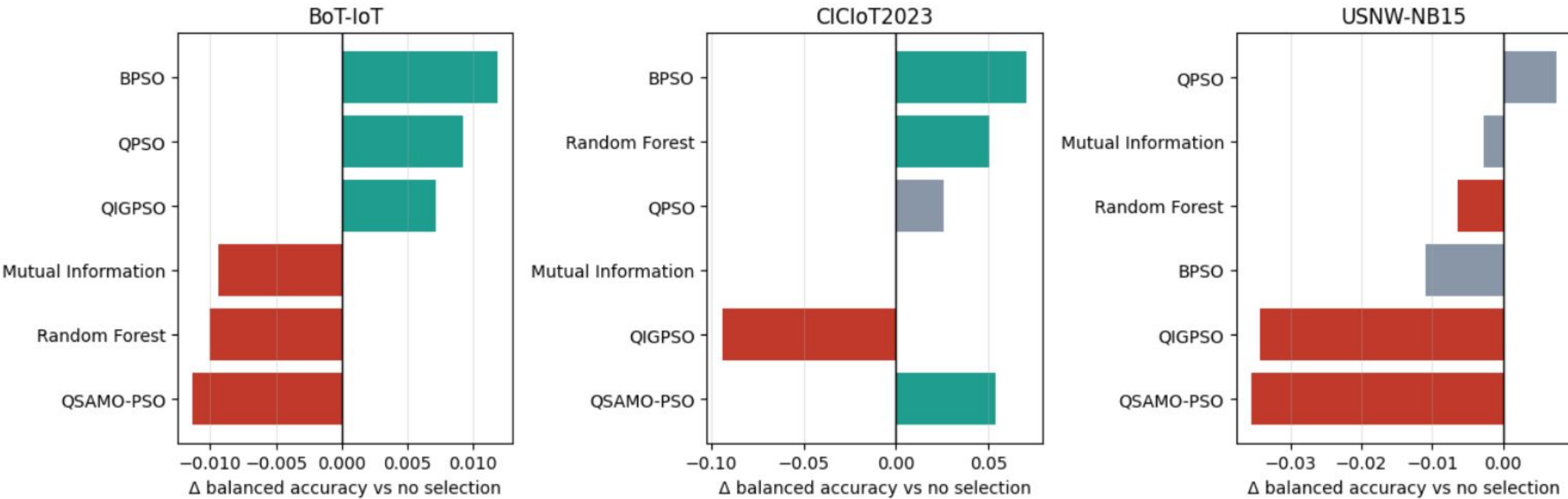


Illustrates opimal solutions (similar colours) that do not compromise conflicting elements of the multi-objective function.

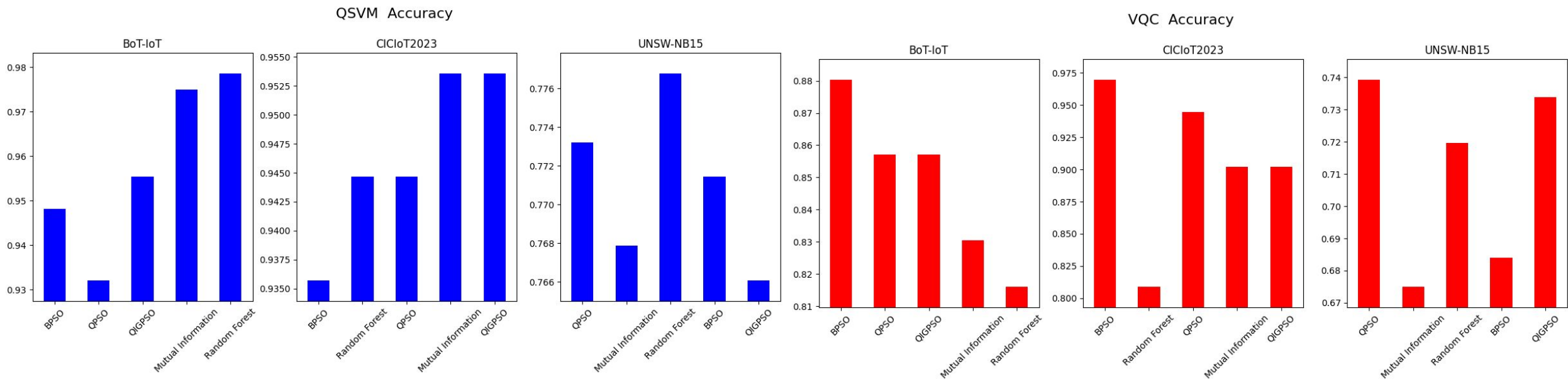
Tailored Feature Selection Helps Improve Accuracy

Decision Tree models were used on all datasets

Search-based selection: real, significant wins on ALL datasets



Success on Quantum Models



Success on classical models did not correlate with success with quantum models



TEAM C

Quantum Training, Robustness & Honest Limits

Arjun E Naik · La Wun Nannda

Training Process

FULL TRAINING PIPELINE

- Team A prepared data splits (train / cal / test / zeroday) + Team B selected features
- Data Scaling
- PCA to 95% variance → set the number of qubits
- Build variational circuit
 - angle encode
 - data reupload
 - depolarizing noise
- Train model
- Save model checkpoint

2 ARCHITECTURES

- Different loss functions
- MAQT architecture:
 - margin-aware
 - loss function = $L_{CE} + (\text{Lambda1} * L_{intra}) + (\text{Lambda2} * L_{inter})$
- VQC architecture:
 - loss function = L_{CE}

3 DATASETS

- BoT-IoT
- CICIoT2023
- UNSW-NB15

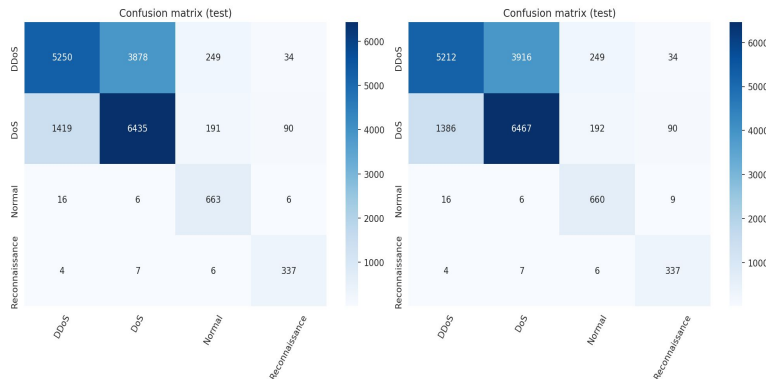
Training Settings

- **PennyLane:** `default.mixed` backend
- **Learning Rate:**
 - Fixed: 0.05, 0.03, 0.01, 0.005
 - Decaying: start at a higher LR (0.01), then smoothly lower it in a cosine curve toward a small floor
- **Epochs with Early Stopping:**
 - Train for up to 50 epochs
 - If validation loss doesn't improve for several epochs, stop
- **Reuploading layers:** 2-3
- **Seeds:** only 42
- **Lambdas:**
 - (0.5, 0.1) for BoT-IoT
 - (0.5, 0.66) for CICIoT2023
 - (0.66, 0.33) for UNSW-NB15

Known-Class Test Accuracy Results

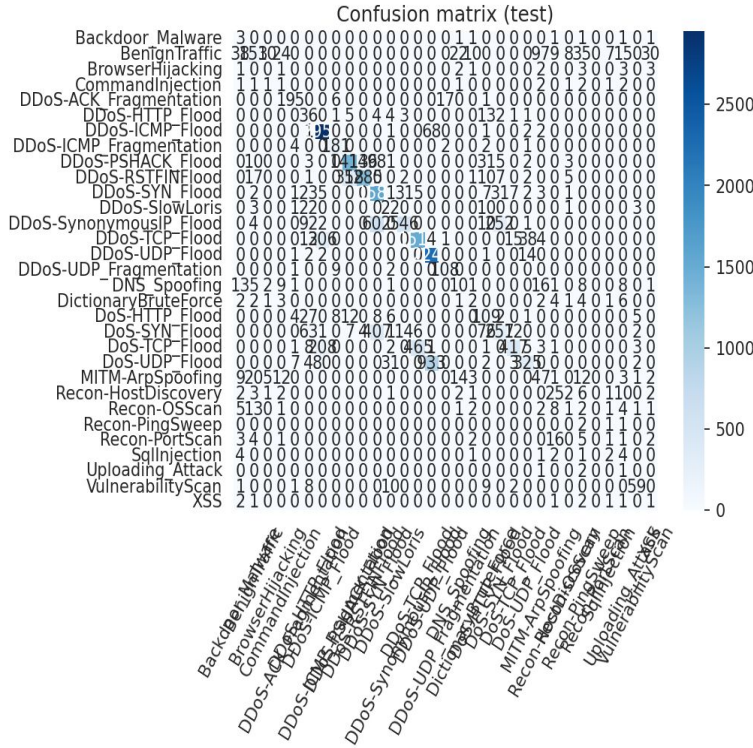
BoT-IoT

- MAQT accuracy: 68% on full test split
- VQC accuracy: 68% on full test split



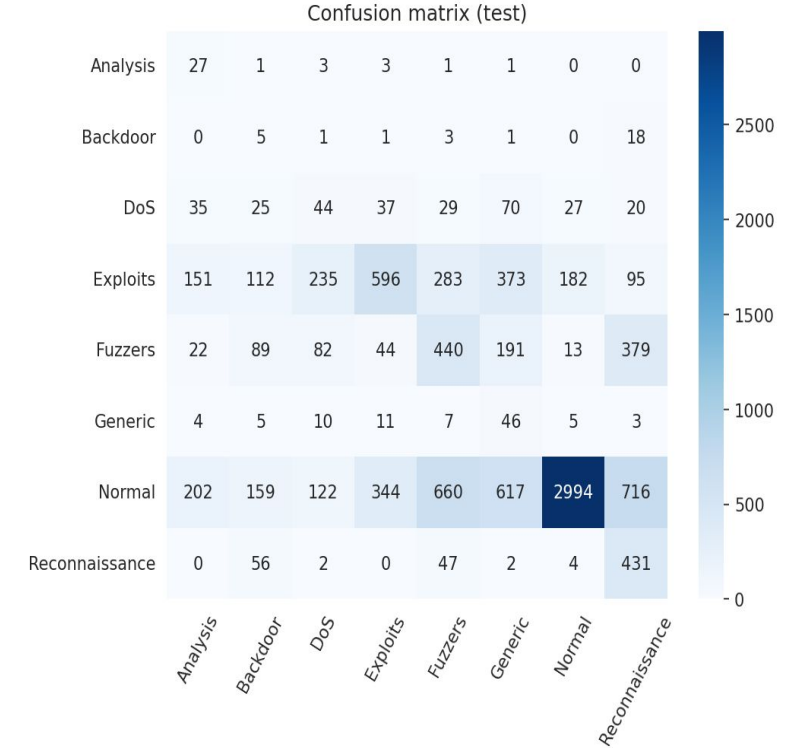
CICIoT2023

- MAQT accuracy: 71% on full test split
- VQC accuracy: 71% on full test split

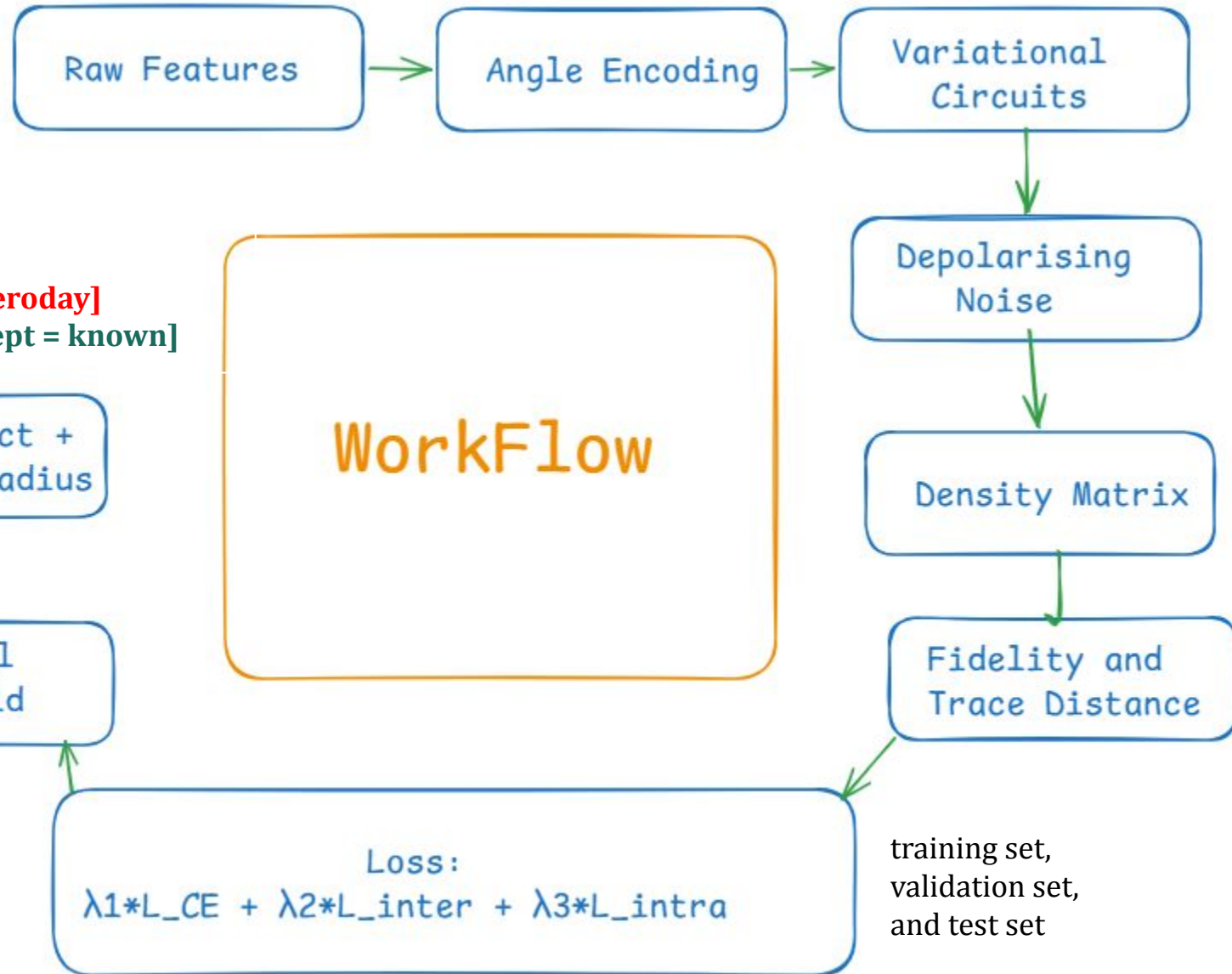


UNSW-NB15

- MAQT accuracy: 45% on full test split
- VQC accuracy: 45% on full test split



Training set



Accept/Reject rule:

- if score > threshold q -> [reject = zeroday]
- else if score <= threshold q -> [accept = known]

zeroday set

calibration set

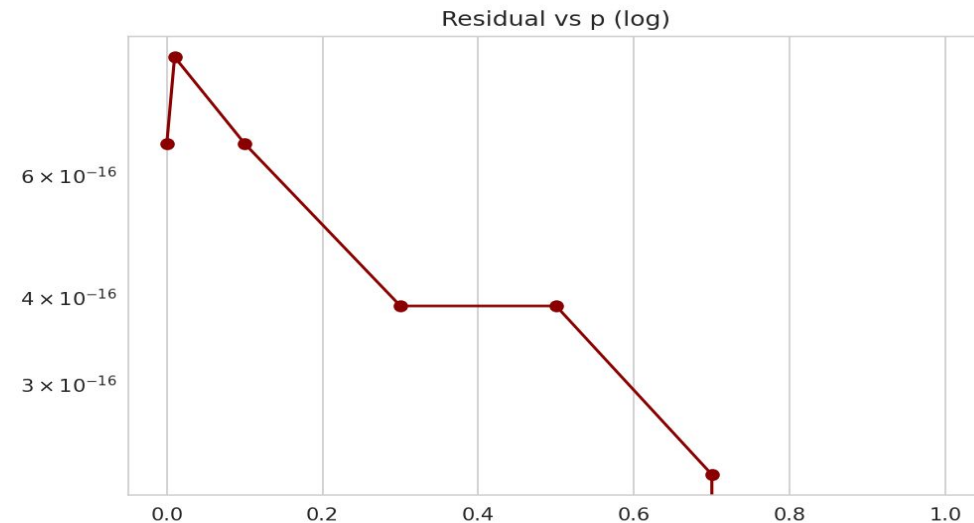
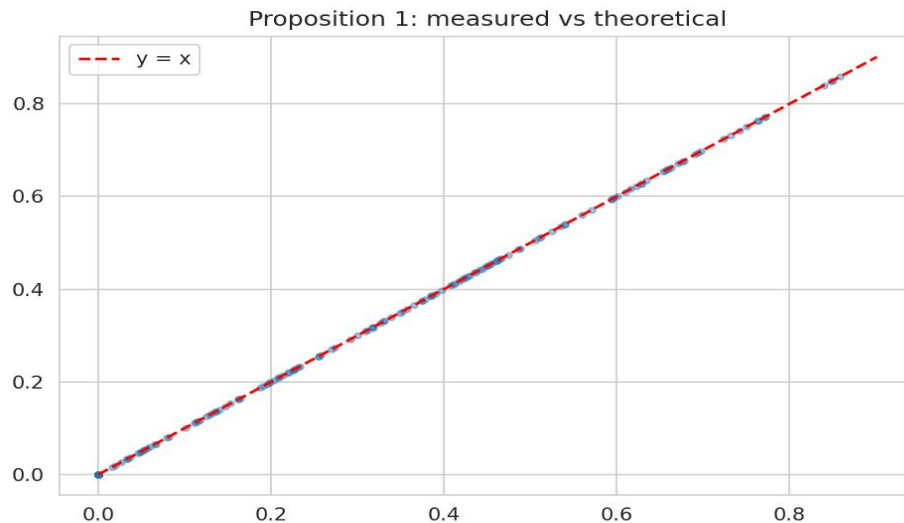
training set,
validation set,
and test set

Noise Identity to Machine Precision

Proposition 1 (Noise contraction). *For the depolarizing channel Λ_p and any two states ρ, σ ,*

$$D_{\text{tr}}(\Lambda_p(\rho), \Lambda_p(\sigma)) = (1 - p) D_{\text{tr}}(\rho, \sigma). \quad (2)$$

In words: depolarizing noise shrinks the distance between any two states by exactly the factor $(1 - p)$, uniformly. It does not distort the geometry—it rescales it.



Certified Separation (Theory)

Proposition 2 (Adversarial–novelty separation). *Under assumptions (A1)–(A5), for any adversarial perturbation δ with $\|\delta\|_2 \leq \varepsilon$ there exists a rejection threshold τ such that every adversarially perturbed known sample is still accepted and every zero-day sample is still rejected, provided*

$$\varepsilon < \frac{(1 - F_{\text{out}}) - \sqrt{1 - F_{\text{in}}^2}}{2(1 - p)L_\varphi} =: \varepsilon^*. \quad (4)$$

Any threshold τ in the interval

$$\left(\sqrt{1 - F_{\text{in}}^2} + (1 - p)L_\varphi \varepsilon, \quad (1 - F_{\text{out}}) - (1 - p)L_\varphi \varepsilon \right)$$

works, and assumption (A5) together with $\varepsilon < \varepsilon^*$ is exactly the condition that makes this interval non-empty.

(A1) The encoder is Lipschitz.

There exists a constant L_φ such that for all inputs x, x' ,

$$D_{\text{tr}}(|\psi(x)\rangle\langle\psi(x)|, |\psi(x')\rangle\langle\psi(x')|) \leq L_\varphi \|x - x'\|_2.$$

(A5) The separability precondition.

The distance gap is positive:

$$\Delta := (1 - F_{\text{out}}) - \sqrt{1 - F_{\text{in}}^2} > 0.$$

(A2) Known samples are concentrated.

Every known-class sample satisfies $F_{\text{max}}(x) \geq F_{\text{in}}$: each known sample is at least this close in fidelity to its nearest prototype.

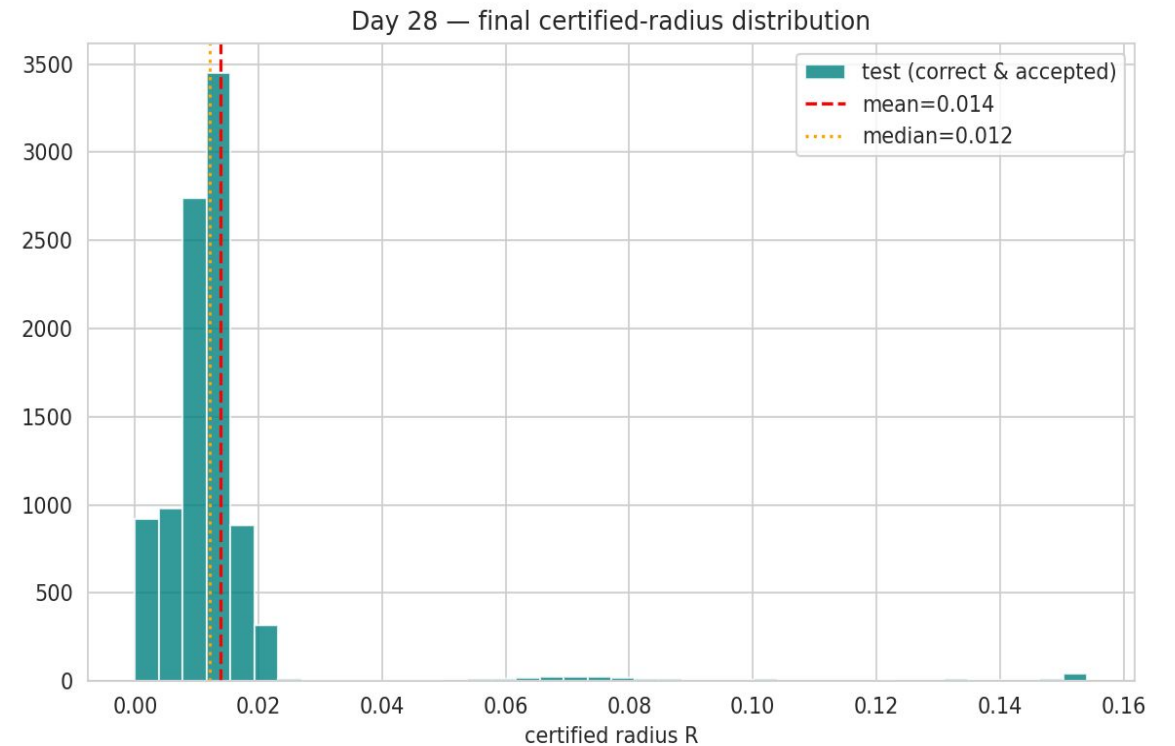
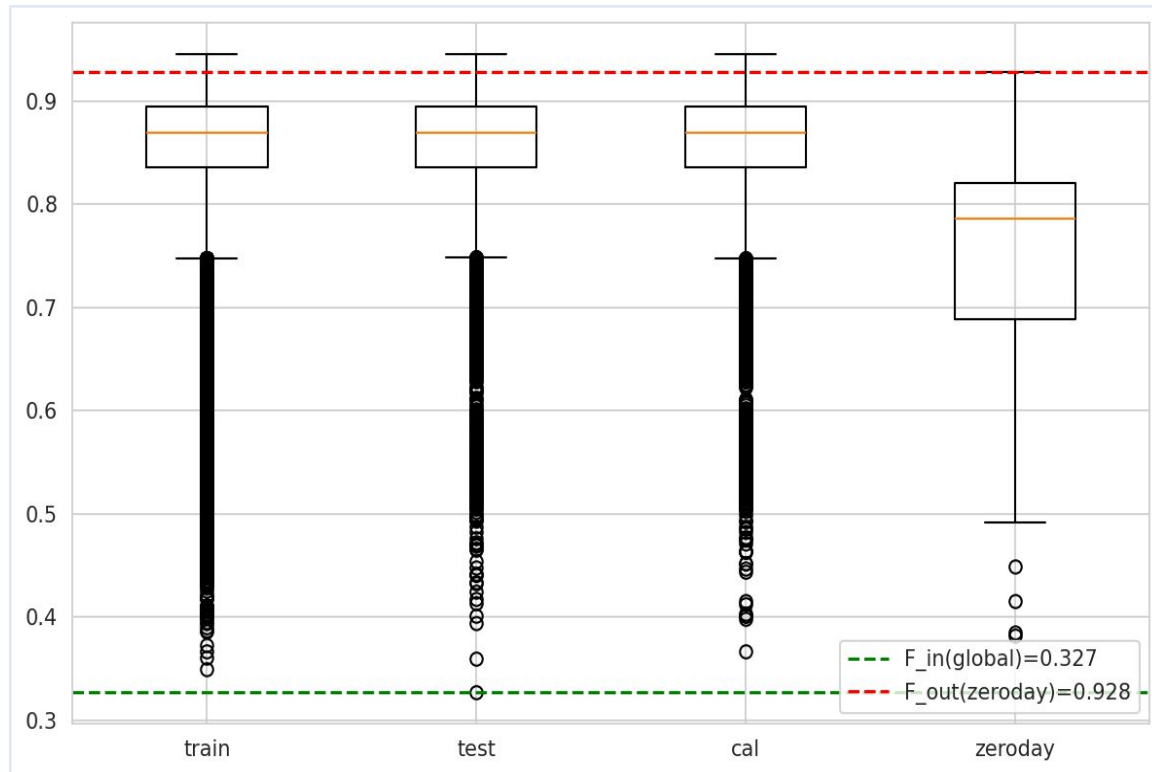
(A3) Zero-day samples are separated.

Every zero-day sample z satisfies $F_{\text{max}}(z) \leq F_{\text{out}}$, with $F_{\text{out}} < F_{\text{in}}$.

(A4) Where the fidelities are measured.

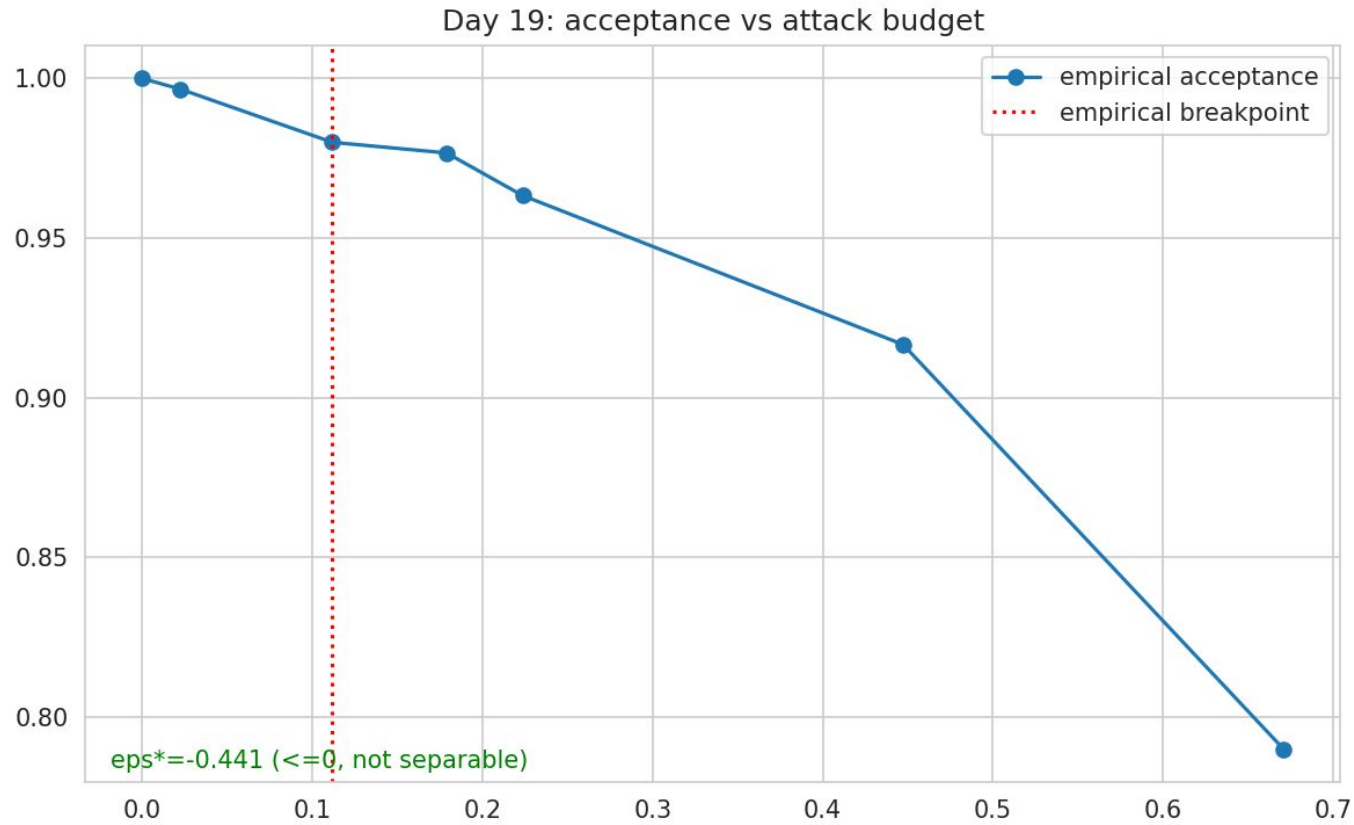
F_{in} and F_{out} are measured on the *post-channel* states $\rho(x)$, at the same noise level p used at inference—the same states the detector actually sees. This matters; see Section 3.4.

Empirical Robustness (BoT-IoT MAQT Results)

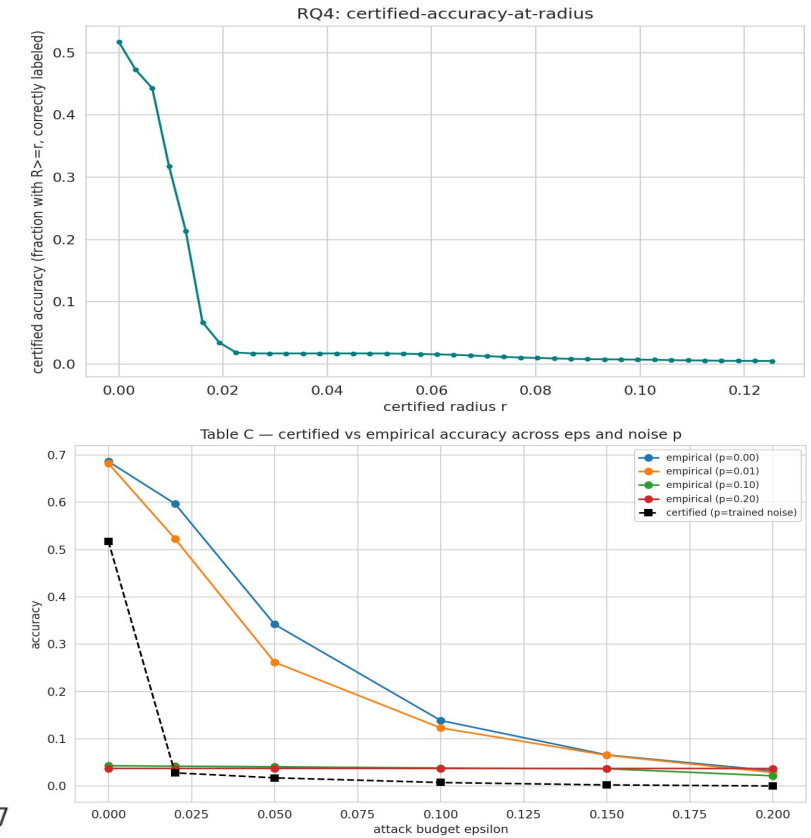


- certified separation is vacuous on real traffic (small fidelity gap)
- known samples and zero-day samples look too similar to the class prototypes

Empirical $\gg$ certified (sufficient $\neq$ necessary)



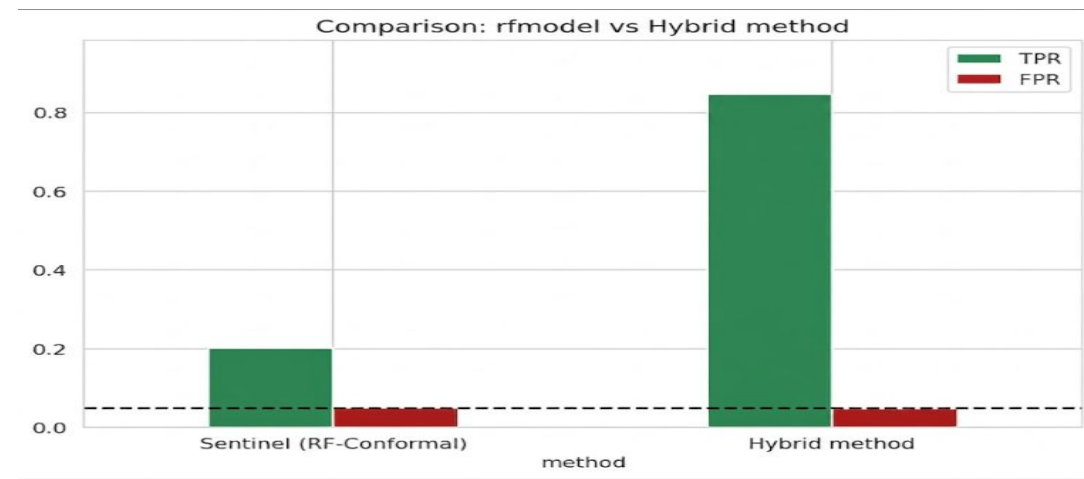
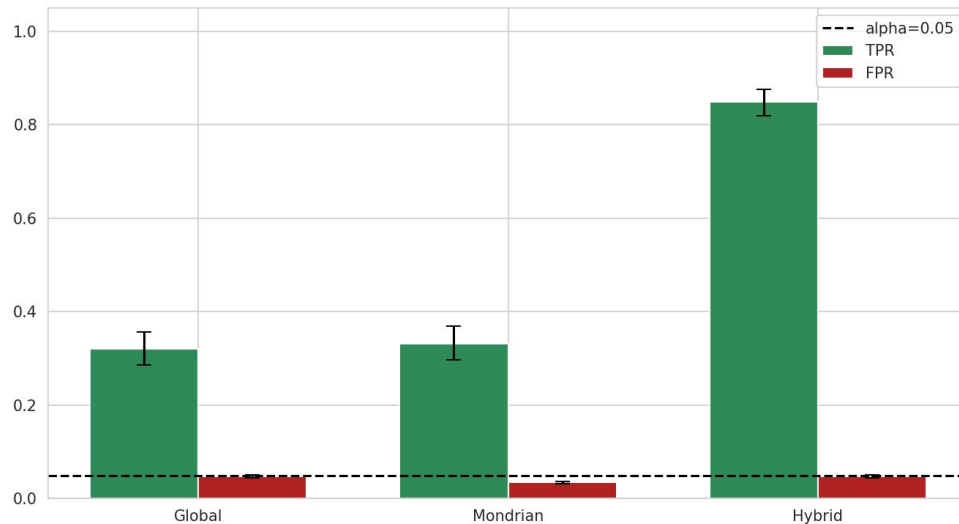
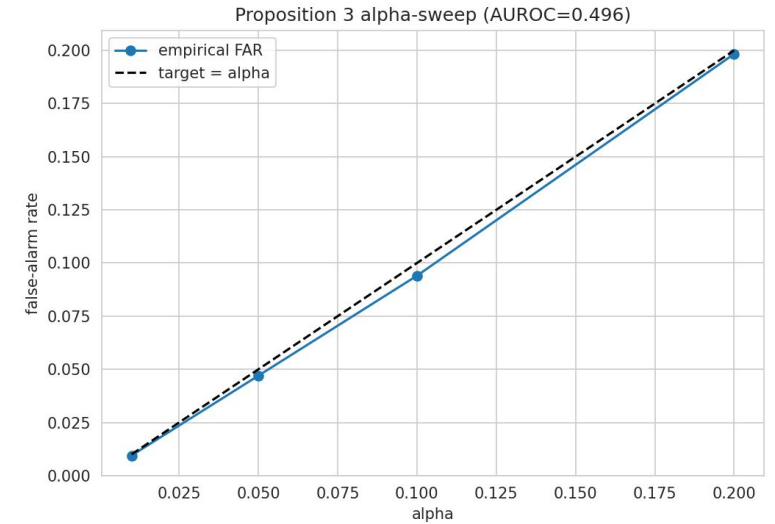
- BoT-IoT MAQT results



Zero-Day Detection: Global q vs Mondrian q vs Hybrid q

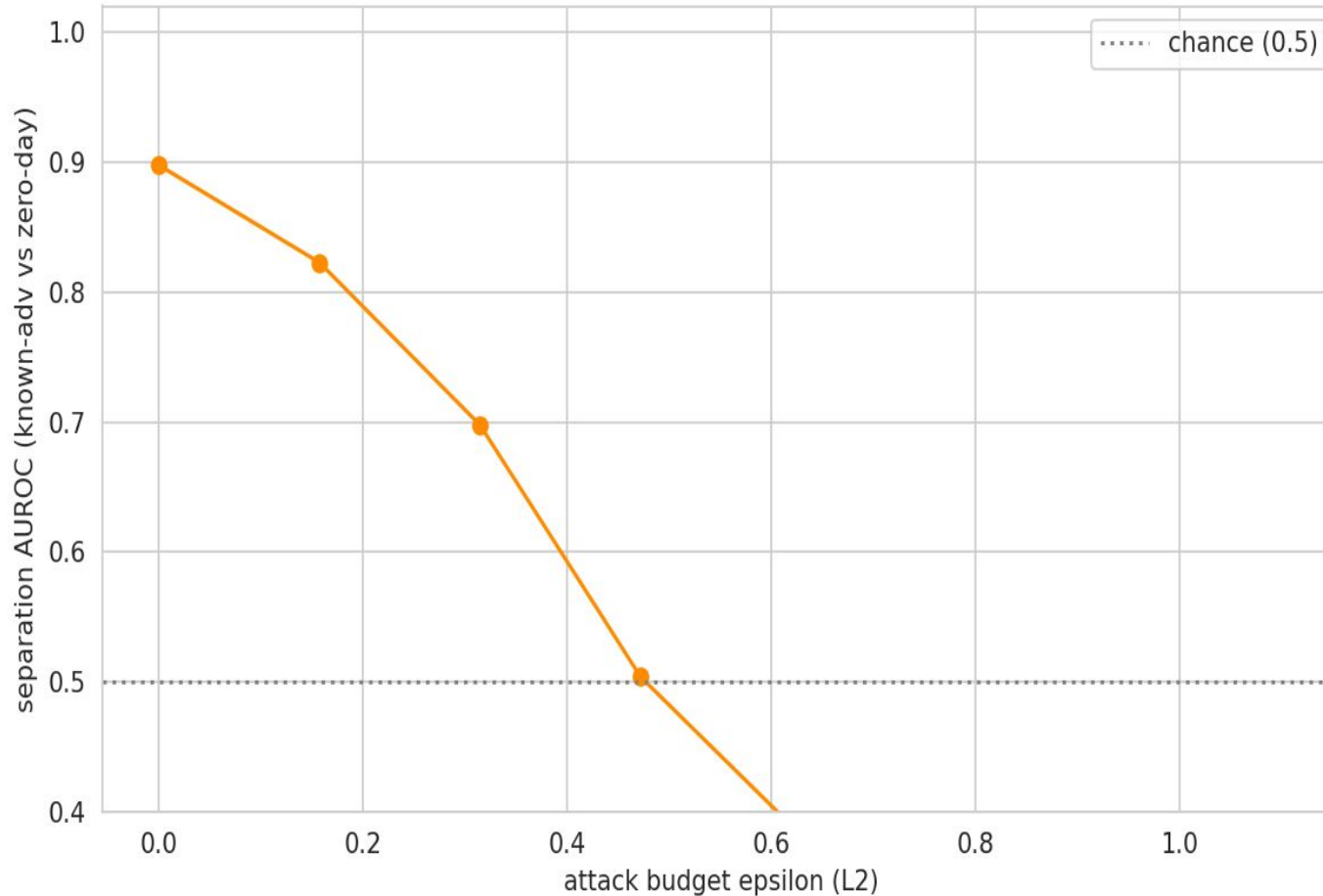
Proposition 3 (Conformal coverage). Let $\{s_1, \dots, s_n\}$ be the nonconformity scores of the calibration set, where $s(x) = 1 - F_{\max}(x)$, and let $q = s_{(k)}$ with $k = \lceil (1 - \alpha)(n + 1) \rceil$ be the k -th smallest calibration score. If the calibration scores and the score of a fresh known-class sample X are exchangeable, then

$$\mathbb{P}(s(X) > q) \leq \alpha. \quad (5)$$



Adversarial Perturbation Destroys Separation

Figure 3 — Disentanglement: separation AUROC vs attack budget



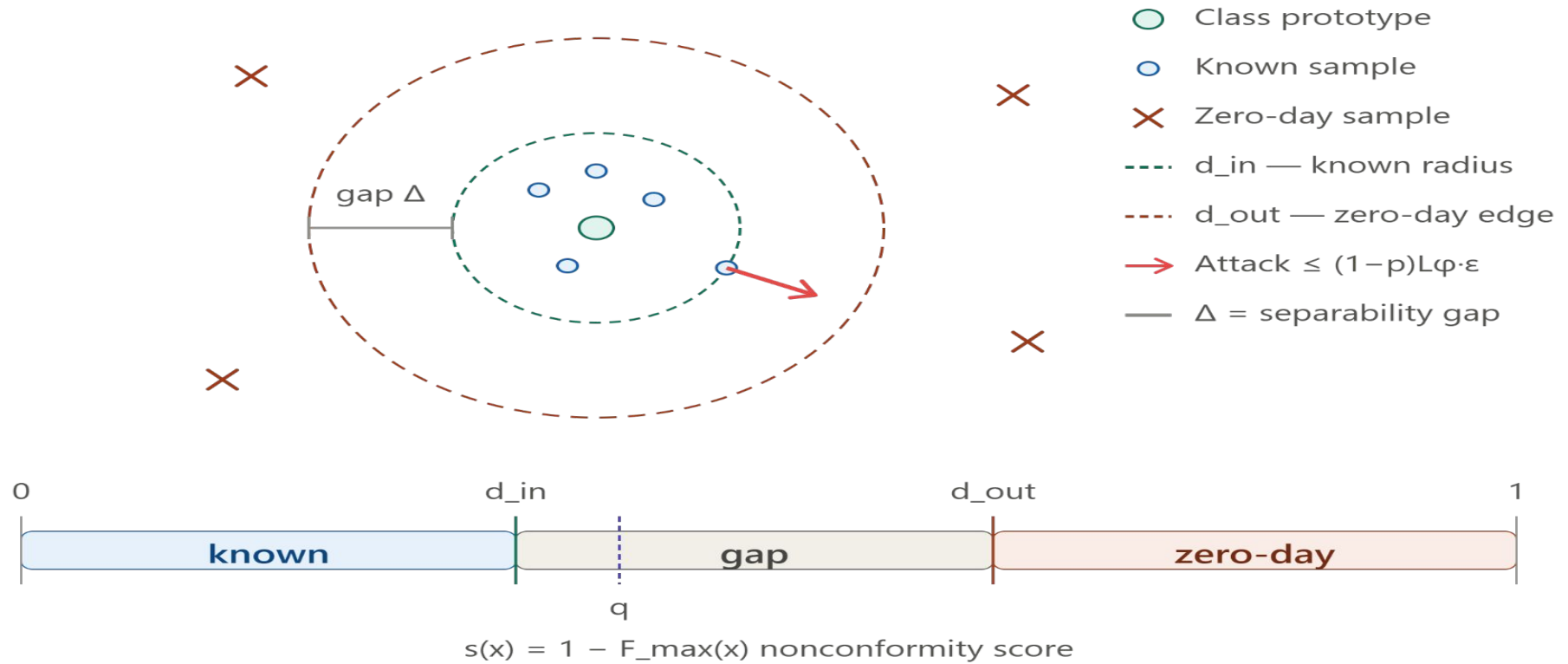
- The empirical curve confirms that separation degrades continuously and collapses around $\epsilon \approx 0.45$.
- The theory and the experiment agree on the outcome: the geometry **does not support** certified separation.

Trust Region Visualized



- No clean separation
- Distribution overlap
- Zeroday Curved manifold cut through known class regions rather than occupying outside of the class

MAQT Prototype Visualization



Separability gap Δ
 $\Delta = d_{out} - d_{in}$

Attack budget ϵ^*
 $\epsilon^* = \Delta / (2(1-p)L\phi)$

Certified radius R
 $R = m / (2(1-p)L\phi C_f)$

A1 Encoder is Lipschitz

A2 Known: $F \geq F_{in}$

A3 Zero-day: $F \leq F_{out}$

A4 Measured at same noise p

A5 Requires gap $\Delta > 0$

Prop 1: depolarizing noise contracts every distance by exactly $(1-p)$

Theory vs. Empirical

Component	What Theory Predicts	What the Experiment Shows
Prop 1 — Noise contraction	Exact identity: D_{tr} shrinks by $(1 - p)$	Residual = 8.88×10^{-16} (machine precision)
Lemma 1 — Lipschitz bound	$L_\phi = 1.0$ is an upper bound	Max sampled ratio = 0.7315
Prop 3 — Conformal coverage	Empirical $FAR \leq \alpha$	FAR tracks α perfectly; exch. AUROC = 0.496
A2 — Known concentration	Known samples near prototypes (F_{in} high)	Worst-case $F_{in} = 0.327$ (heavy tail)
A3 — Zero-day separation	Zero-days alien to all prototypes (F_{out} low)	$F_{out} = 0.928 \gg F_{in} = 0.327$
A5 — Separability precondition	$\Delta = (1 - F_{out}) - \sqrt{(1 - F_{in}^2)} > 0$	$\Delta = \mathbf{-0.87}$ (strict), $\mathbf{-0.80}$ ($\beta=0.05$)
Prop 2 — Certified budget ε^*	Positive ε^* guarantees separation	$\varepsilon^* = \mathbf{-0.44}$ (vacuous, no safe budget)
Detection (quantum only)	Not guaranteed by Prop 2 alone	TPR = 32% , FPR = 4.7%
Certified radius	R bounds robust region	Mean $R = \mathbf{0.014}$ (microscopic)
Hybrid (QS-Net + IsoForest)	Zero-day TPR	TPR = 85%, FPR = 4.8%

Limitations

- **Time Constraints:** Experimentation was limited by the tight schedule due to:
 - Qubit Count: Exponentially increases training time per epoch.
 - Epoch Count: Linearly increases overall training duration.
- **Hardware Constraints:** Training requires continuous GPU access, but free-tier cloud platforms have daily or weekly limits.

Future Work

- Improve accuracy (more hyperparameter tuning, different lambdas, etc.)
- Run more epochs upto where it converges the local minima
- Streaming / adaptive conformal calibration
- Experiment with more qubits

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THANK YOU

Questions are welcome